

Exercise 6 – Design for Manufacturing (DFM) Optimization

Comparison of Original and Redesigned Sheet-Metal Enclosure

Engineering Design Analysis

SOLIDWORKS sheet-metal redesign focused on manufacturability, material efficiency, weight reduction, ventilation and fabrication simplicity.

1. Original Design – Observed Characteristics

The original enclosure is a basic sheet-metal construction formed from a base panel and surrounding bent flanges. The screenshots show a compact enclosure with mounting openings and folded side/rear walls. The design is functional and relatively simple, but it provides less visible opportunity for material optimization and ventilation.

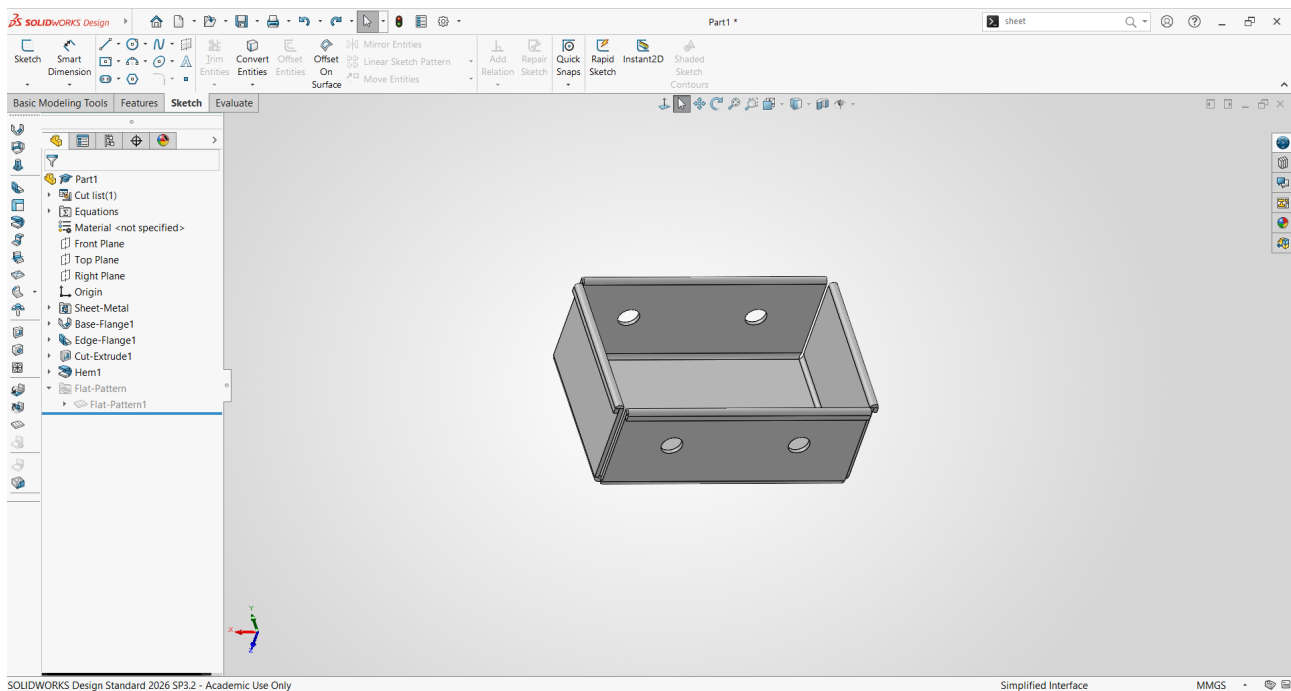


Figure 1 – Original Design

Key features visible in the original

- Basic sheet-metal enclosure structure.
- Base flange forming the main supporting surface.
- Bent edge/side flanges forming the enclosure walls.
- Mounting holes/openings integrated into the enclosure.
- Folded geometry suitable for sheet-metal fabrication.

Design assessment: The original design establishes the required enclosure form, but its visible geometry is comparatively plain and does not show the added ventilation and weight-reduction features introduced in the redesign.

2. Redesigned / DFM-Optimized Design

The redesigned enclosure keeps the basic sheet-metal architecture while introducing features intended to improve practical manufacturing and functional performance. The SOLIDWORKS feature tree visibly includes Base-Flange, Edge-Flange, Cut-Extrude, Corner Relief, a mirrored feature and Flat-Pattern capability.

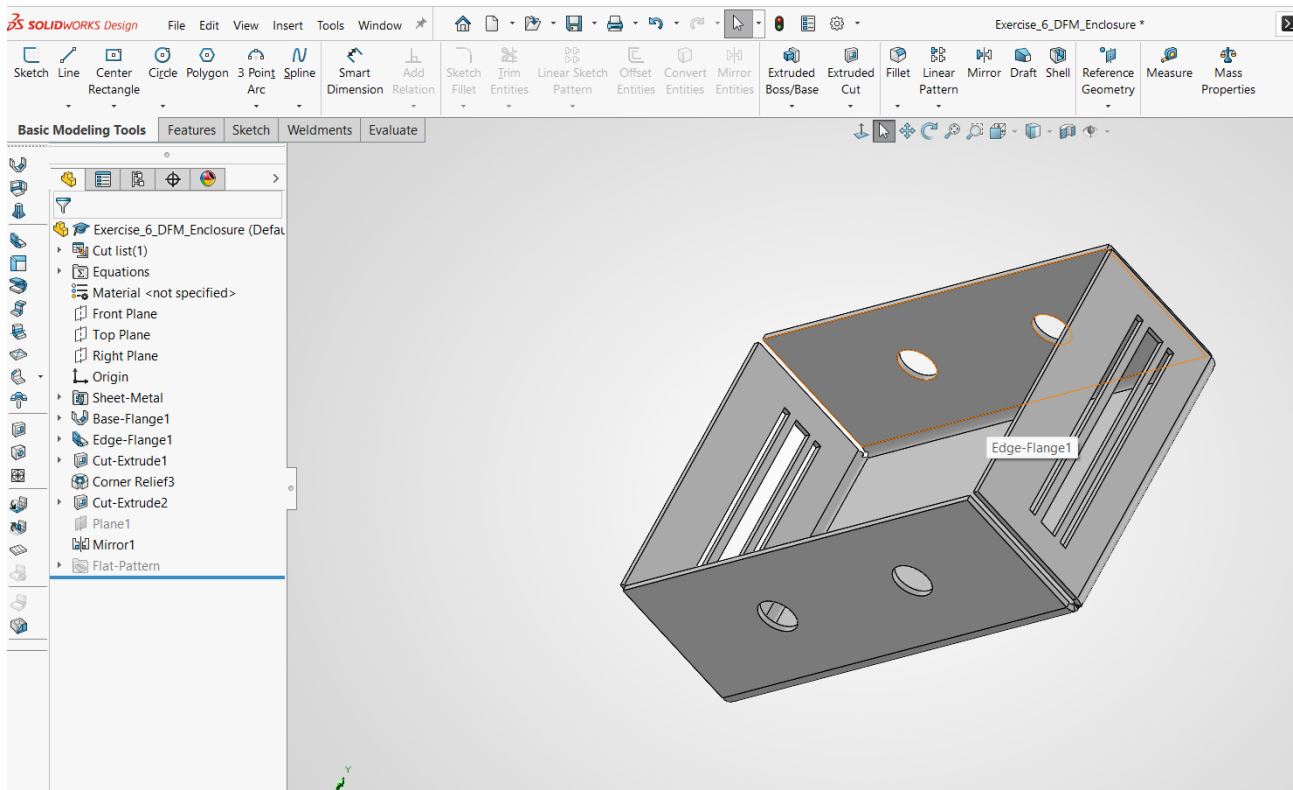


Figure 2 – Redesigned DFM-Optimized Design

Clearly visible improvements

- Added elongated cut/slot features on the side walls.
- Mirrored slot geometry provides consistent features on opposite sides.
- Corner relief is incorporated into the sheet-metal design.
- Mounting holes remain integrated into the enclosure.
- Flat-pattern capability supports fabrication directly from sheet stock.
- The redesign maintains a coherent bent sheet-metal construction rather than adding separate bulky components.

Why this is better: The redesign does more than change the appearance. The slots remove material where it is not needed for a solid wall, provide a ventilation path, and can reduce part mass. Corner relief and a proper flat pattern support cleaner sheet-metal forming and reduce the risk of interference at bends.

3. DFM Improvement Analysis

Material and weight

The ventilation/weight-reduction slots remove portions of sheet material while retaining the surrounding enclosure walls. This can reduce raw-material consumption and part weight.

Manufacturing

The redesign remains compatible with a sheet-metal workflow: flat-pattern cutting followed by bending and finishing. The use of repeated/mirrored slot geometry can also make the design more consistent for CNC laser, plasma or similar sheet-cutting processes.

Ventilation and functionality

The added slots create openings in the side walls, improving airflow compared with a fully closed side wall. This is particularly useful when the enclosure contains components that generate heat.

Bending and corner quality

Corner relief is a sheet-metal DFM feature that provides clearance at intersecting bends. It helps avoid material interference and supports cleaner forming of the enclosure.

Assembly and inspection

The redesign keeps the mounting-hole concept while providing more open side geometry. This can improve access for inspection and handling.

4. Manufacturing Approach

Original design – likely workflow

Sheet stock → profile/feature cutting → hole operations → bending → finishing → assembly

The original enclosure already benefits from sheet-metal fabrication because several walls are formed by bends rather than assembled as separate solid blocks.

Redesigned design – optimized workflow

Sheet stock → flat-pattern cutting with integrated holes/slots → bending → finishing → assembly

The redesigned model makes the cutting stage more feature-rich by incorporating ventilation/weight-reduction slots and corner relief into the sheet-metal part. The flat pattern provides a direct representation of the developed sheet for fabrication.

5. Final Engineering Evaluation

Category	Original	Redesigned
Manufacturability	Good	Very Good
Material efficiency	Fair	Very Good
Weight optimization	Fair	Very Good
Cost efficiency	Good	Very Good
Assembly	Good	Very Good
Ventilation	Fair	Excellent
Design simplicity	Good	Very Good
Overall DFM suitability	Good	Very Good

6. Overall Design Improvement

The redesigned enclosure is a stronger DFM-oriented solution because it retains the useful sheet-metal structure while adding purposeful features instead of simply increasing part complexity. The most important improvement is that the new features serve multiple functions: the slots can reduce unnecessary material while also providing ventilation; the mirrored layout improves consistency; corner relief supports sheet-metal forming; and the flat pattern provides a fabrication-ready representation.

7. Conclusion

The redesign demonstrates a practical application of Design for Manufacturing principles to a sheet-metal enclosure. Instead of replacing the basic structure, the design improves it by introducing material-removal slots, corner relief, repeated geometry and a usable flat pattern.

From the visible model, the redesigned part offers better ventilation, a more material-efficient geometry and features that are more deliberately aligned with sheet-metal fabrication. These changes have the potential to reduce material usage, part weight and manufacturing effort while maintaining the enclosure's basic mounting and structural functions.